BIT 319 CAT 1:

Group Activity: MIS and Decision-Making

Task

- Work in groups of 7–9 students.
- Discuss how MIS can help in your assigned/selected sector (listed below) (e.g., Hospitals, Universities, Banking, Agriculture, etc.).
- **Guiding Questions**

i. How can MIS support everyday operations (operational level)?

ii. How can MIS help with medium-term planning (tactical level)?

iii. How can MIS guide long-term strategy (strategic level)?

iv. How does MIS reduce uncertainty or risks in this sector?

v. What decision-support tool or technique (e.g., dashboards, simulations, forecasting) could be applied?

- **Expected Output**

i. A clear list of MIS contributions for your sector.

ii. Examples of decision-making levels supported by MIS.

iii. At least one decision-support technique identified.

iv. A presentation summary (about 15 slides).

v. Active participation from all members — everyone will take part in the final presentation.

Each group to select one sector to work on from the list below

1. **Education Technology (EdTech & E-Learning Providers)**

- Online learning platforms management
- Student performance tracking & analytics
- Personalized learning systems (adaptive learning)
- Digital resource management (ebooks, simulations, labs)
- Cybersecurity in e-learning systems

- Integration with university/school MIS

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1. **Health Insurance and Social Security**

- Claims processing and fraud detection
- Customer data management
- Premium pricing and risk assessment
- Decision support for policy approvals
- Predictive analytics for healthcare costs
- Compliance with government regulations

2. Banking and Financial Institutions

- Loan approval & risk assessment
- Fraud detection & prevention
- Customer relationship management
- Online/mobile banking services
- Investment decision support
- Regulatory compliance reportin

3. **Manufacturing and Supply Chain**

- Inventory management
- Production planning & scheduling
- Quality control and monitoring
- Supply chain integration (ERP/SCM)
- Cost management & waste reduction
- Predictive maintenance of machines

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4. **Retail and E-Commerce**

- Customer behavior analytics
- Pricing and discount optimization
- Inventory replenishment
- Personalized marketing
- Payment systems & transaction security
- Demand forecasting

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5. **Agriculture and Agribusiness**

- Farm resource allocation (water, fertilizer, labor)
- Market information systems for pricing
- Supply chain management (farm to market)
- Weather forecasting & crop planning
- Pest and disease monitoring systems
- Financial services for farmers (loans/insurance)

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6. **Transport and Logistics**

- Route optimization & scheduling
- Fleet management systems
- Cargo tracking & monitoring
- Safety and compliance management
- Real-time traffic management
- Cost reduction in logistics operations

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7. **Government and Public Administration**

- E-government services (Huduma Kenya–style)
- Tax collection and compliance (like iTax)
- National ID/Passport management systems
- Social services distribution (cash transfers)
- Electoral management systems
- Policy decision support with big data

8. **Energy and Utilities**

- Power generation forecasting
- Load balancing and grid management
- Billing and revenue collection
- Predictive maintenance of infrastructure
- Renewable energy monitoring
- Customer complaint management

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9. **Tourism and Hospitality**

- Reservation and booking systems
- Customer feedback and sentiment analysis
- Demand forecasting (peak/off-peak)
- Staff scheduling & resource allocation
- Loyalty program management
- Risk management in tourism operations

To be submitted/presented on 30th of September 2025